

CS6765 NLP: Research Report

A Review of “GraphNarrator: Generating Textual Explanations for Graph Neural Networks”

Your Name

Student ID: 00000000

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1 Summary

1.1 Motivation

Graph Neural Networks (GNNs) are widely adopted for tasks where relationships between entities matter, such as molecular property prediction, financial fraud detection, community analysis in social networks, and recommendation systems. Although these models achieve impressive accuracy, their internal decision processes are notoriously opaque. Users often only see the final prediction, with no clear understanding of how different parts of the graph influenced the outcome. This lack of transparency can limit trust, hinder debugging, and make GNNs inappropriate for sensitive domains where justification is essential.

Existing GNN explanation methods mostly rely on saliency-based visualizations that highlight important nodes or edges. While these methods are useful for experts, they remain unintuitive for general users. Visual or structural explanations do not communicate full reasoning and often require domain knowledge to interpret correctly. The gap between low-level graph saliency and high-level human reasoning motivates the need for explanations that “speak the user’s language.”

Natural language explanations offer a promising bridge, but generating faithful textual descriptions from GNNs is challenging. The main difficulty lies in translating structural graph information—non-linear, non-sequential, and topology-driven—into coherent language that reflects the GNN’s actual reasoning. Previous attempts to use LLMs for explanations often suffer from hallucination, overgeneralization, or lack of grounding. Motivated by these challenges, the authors propose GraphNarrator, a framework that aims to produce explanations that are both accurate and genuinely reflective of the GNN’s decision-making process.

1.2 Contribution

The paper introduces **GraphNarrator**, a novel system that integrates saliency extraction, iterative refinement, and model distillation to generate faithful natural language explanations for graph predictions. Its key contributions include:

- **A structured transformation pipeline** that converts raw graph saliency signals into “saliency paragraphs”—clear textual summaries that describe which graph substructures influence a prediction and why.
- **An expert-iteration refinement framework** that evaluates candidate explanations using criteria designed specifically for GNN interpretability: input faithfulness, output alignment, and conciseness.
- **A distillation-based explainer model** that learns to produce final explanations in a single step, greatly improving inference speed and making the method deployable in real-world scenarios.
- **Comprehensive evaluation** through both automatic metrics and human assessment on three Tree-annotated Graph (TAG) datasets, showing improvements in factual grounding, readability, and human preference over existing baselines such as GNNExplainer and LLM prompting methods.

Overall, the paper contributes a unique and effective approach for bridging symbolic reasoning from graphs with textual explanation techniques from NLP.

1.3 Methodology and/or Argument

The GraphNarrator framework consists of three major stages. First, the system extracts saliency information from a trained GNN to determine which nodes and edges are most influential for the prediction. Traditional saliency maps are difficult for end-users to interpret, so the authors verbalize them into *saliency paragraphs*. These paragraphs describe important substructures, their contextual roles, and their contribution to the model’s decision.

Second, the expert-iteration process refines these explanations. In each iteration, an LLM produces an explanation that is then evaluated by a critic model across three dimensions. *Input faithfulness* ensures the explanation reflects the structural evidence identified by the saliency method. *Output faithfulness* ensures the explanation stays aligned with the predicted class instead of inventing new justifications. *Brevity* encourages concise communication, reducing redundancy and improving readability.

The LLM receives feedback and generates updated explanations, gradually improving clarity and grounding. This iterative approach resembles reinforcement learning or human-guided critique loops commonly used in modern NLP, but adapted specifically for graph explanations.

In the final stage, the entire multi-step pipeline is distilled into a single explainer model. The distillation phase trains the LLM to imitate the high-quality explanations produced during expert iteration, enabling one-step inference. This greatly reduces computational cost and removes the need for repeated refinement during deployment.

The authors argue that this combination—grounding via saliency, refining via expert feedback, and simplifying via distillation—provides a robust methodology for generating faithful, human-readable explanations for GNNs.

1.4 Conclusion

The evaluation demonstrates that GraphNarrator consistently produces more accurate, concise, and understandable explanations than existing graph explanation methods. Automatic metrics show

improved alignment between the explanations and the model’s saliency signals. Human evaluators also prefer GraphNarrator’s explanations for clarity, relevance, and usefulness.

The authors highlight that natural language explanations can significantly enhance user trust in GNN-based systems, particularly when the domain requires strong justification. They also note limitations: reliance on LLMs increases computational cost, and experiments are limited to moderately sized datasets. Nonetheless, GraphNarrator presents a strong foundation for integrating natural language reasoning with graph-based learning models and opens multiple directions for future exploration.

2 Critique

2.1 Strengths

One of the strongest aspects of the paper is the clear motivation and the practical relevance of the problem it addresses. Explainability is increasingly important as machine learning models are deployed in sensitive areas such as healthcare, legal systems, and financial analysis. By focusing on GNNs—a model family with real-world impact—the authors tackle a timely and meaningful challenge.

Another strength is the use of a structured iterative refinement process. Instead of relying on ad hoc prompting or single-pass LLM generation, the authors use systematic criteria to evaluate and improve explanations. This methodology differentiates GraphNarrator from prior work and helps ensure the generated text is grounded, concise, and aligned with the model’s internal reasoning.

The distillation step is also particularly impressive. Many explainability techniques are difficult to deploy in practice because they require multiple passes or complex training pipelines. The fact that the authors compress the entire iterative process into a single efficient model demonstrates attention to usability and scalability—qualities that many academic explanation systems lack.

2.2 Weaknesses and Limitations

The main limitation of the approach is its dependence on the quality of saliency maps. Saliency methods for GNNs are known to be noisy and sometimes inconsistent. If the saliency extraction is inaccurate, even a well-written explanation will be misleading. Although the expert-iteration process helps mitigate this, it cannot fully compensate for an unreliable foundation.

Another limitation is that the method assumes explanations can be fully captured using textual descriptions. Some graph predictions may depend on subtle structural interactions that are inherently difficult to phrase in natural language. In such cases, textual explanations may oversimplify the model’s reasoning. A combination of structured diagrams and text might be more effective.

Finally, the reliance on LLMs introduces concerns about reproducibility and computational accessibility. Training and running LLMs—even distilled ones—can be expensive, which may limit the use of GraphNarrator in resource-constrained settings. The paper would benefit from more discussion on how to adapt the method for lightweight deployments.

3 Synthesis

GraphNarrator’s ideas connect strongly with course topics such as interpretability, natural language generation, alignment, and multimodal reasoning. In particular, it illustrates how explanation principles traditionally associated with NLP—such as rationale extraction or chain-of-thought—can be extended to non-text domains. The method reflects a broader trend in machine learning: combining symbolic or structured signals with the expressive power of LLMs.

The framework also aligns with discussions on model alignment. The expert-iteration loop closely resembles reinforcement learning from human feedback (RLHF) and preference optimization techniques used to align LLMs with human expectations. Instead of aligning to human judgment directly, GraphNarrator aligns explanations to model behavior and structural evidence, which is an interesting twist on standard alignment techniques.

From an NLP perspective, the work raises important questions about how language can serve as a medium for communicating complex model behavior. Similar to explanations in sentiment classification, question answering, or summarization, GraphNarrator demonstrates that natural language can act as an interpretability layer even when the underlying data is not linguistic.

Looking forward, this paper suggests several promising research directions. One avenue involves extending the method to larger and more diverse graph datasets, including molecular graphs, knowledge graphs, and financial transaction networks. Another promising direction is integrating domain knowledge or symbolic reasoning into the explanation process to enhance factual accuracy. Finally, methods from this paper could be adapted for multimodal explanation systems, where text, images, and structured data combine to form richer explanations.

Overall, GraphNarrator represents a strong example of how NLP techniques—particularly LLM-based reasoning—can be used to improve interpretability across different areas of machine learning. It reinforces the course’s central message: language is not only a modeling target but also a powerful tool for understanding and communicating model decisions.

References

- [1] Bo Pan, Zhen Xiong, Guanchen Wu, Zheng Zhang, Yuntong Hu, Yifei Zhang, and Liang Zhao. *GraphNarrator: Generating Textual Explanations for Graph Neural Networks*. Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL), 2025.